Lesson 7 - A City Tour Where Each City Is Visited Exactly Once

7.1 Special graphs

Question 91

Level of difficulty: E

What is meant by a star graph? How many nodes and edges does the star graph S_n have? Draw the star graph S_4.

Question 92

Level of difficulty: E

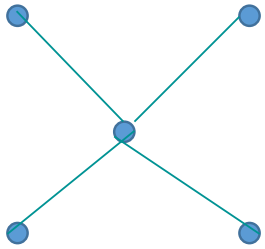
What is meant by cycle graphs? What properties do they have?

Solution question 91

Solution hint: Familiarize yourself with the theory of star graphs.

Short solution: Central node connected to all other nodes; n edges and n+1 nodes

Detailed solution: A star graph S_n has a central node, which is connected to all other nodes by an edge. The other nodes have no neighbors other than the central node. A star graph S_n has n edges and therefore n+1 nodes.



Solution question 92

Solution hint: Familiarize yourself with the theory of cycle graphs.

Short solution: Same number of nodes as edges, regular of degree 2, planar and Eulerian

Detailed solution: A cycle graph C_n with n (in general, $n \geq 3$ is assumed) vertices has as many vertices as edges, all vertices being connected in a (geometric) cycle. All cycle graphs are regular of degree 2, planar and Eulerian.

Question 93

Level of difficulty: M

What is meant by a complete graph? Are all complete graphs Eulerian? How many edges does a complete graph K_n have?

Question 94

Level of difficulty: M

What is meant by a bipartite graph? What properties must it have? Give an example.

Solution question 93

Solution hint: Familiarize yourself with the theory of complete graphs.

Short solution: From each node, an edge leads to every other node. Not all complete graphs are Eulerian. Number of edges: n * (n–1)/2.

Detailed solution: An undirected simple graph with n vertices in which there exists an edge from every vertex to every other vertex is called a complete graph (or complete polygon) K_n. All complete graphs K_n with n odd are Eulerian. The number of all edges of a complete graph is the triangular number n * (n–1)/2.

Solution question 94

Solution Note: Familiarize yourself with the theory of bipartite graphs.

Short solution: The set of its nodes can be split into two disjoint subsets. Prerequisite: At least two nodes and no loop. Example: Cycle graph with even number of nodes

Detailed solution: An undirected graph is bipartite if the set of its nodes can be split into two disjoint non-empty subsets M and N, such that there are only edges between nodes from M and nodes from N, but no edges between nodes within the set M or the set N. A graph whose edge set is the union of two non-empty disjoint sets M and N, such that each edge has one end in M and one end in N, is a bipartite graph. Because of this definition, it is necessary for a bipartite graph to have at least two nodes and no loops (but it can have multiple edges). In other words, a graph is bipartite if its nodes can be colored black and white such that each edge connects a black and a white node (colorable with two colors). The two subsets M and N are also called bipartite. All cycle graphs C_n with even number of nodes are bipartite.

Question 95

Level of difficulty: D

Explain Kuratowski's theorem. What becomes evident on the basis of this theorem?

Solution question 95

Solution hint: Familiarize yourself with the theory. What does planar mean?

Short solution: A graph G is planar if and only if G does not contain a subgraph G' such that K_5 or $K_{3,3}$ is a minor of G'. Due to Kuratowski's theorem, it is obvious that K_5 and $K_{3,3}$ are the smallest non-planar graphs.

Detailed solution: A graph G_1 is called minor of graph G_2 if G_1 is isomorphic to a subgraph of G_2 formed by node merging. Node merging means that two adjacent nodes are merged into one node by removing one edge connecting these two nodes, while keeping all remaining edges. Using Minor's definition, Kuratowski's theorem can be formulated as follows: A graph G is planar if and only if G does not contain a subgraph G' such that K_5 or $K_{3,3}$ is a minor of G'. Given Kuratowski's theorem, it is evident that K_5 and $K_{,3}$ are the smallest non-planar graphs.

7.2 Hamiltonian graph

Question 96

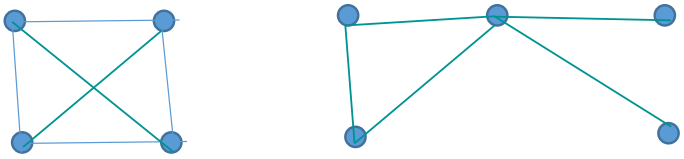
Level of difficulty: E

What is meant by a Hamiltonian path and a Hamiltonian cycle?

Question 97

Level of difficulty: E

Do the following graphs possess a Hamiltonian path and a Hamiltonian cycle?



Solution question 96

Solution hint: Familiarize yourself with the theory of Hamiltonian graphs.

Short solution: The path contains each node only once. Path: not closed. Cycle/circuit: closed

Detailed solution: If an edge path contains each node of a graph exactly once, but is not necessarily closed, then it is a Hamiltonian path. For a directed graph, it is also called a directed Hamiltonian path. A graph that contains a Hamiltonian path is called a semi-Hamiltonian graph. In particular, a Hamiltonian cycle is a closed Hamiltonian path.

Solution question 97

Solution hint: The path contains each node only once.

Short solution: Graph on the left: Hamiltonian path and Hamiltonian cycle; Graph right: no

Detailed solution: Graph on the left: Hamiltonian path and Hamiltonian cycle. There exists a closed path that contains each node only once. Graph on the right: There exists neither an open nor a closed path, which contains each node only once.

Question 98

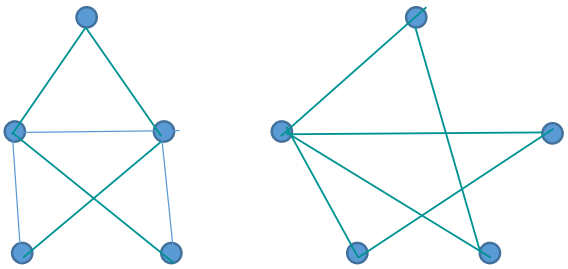
Level of difficulty: M

Are path graphs, cycle graphs, star graphs, and wheel graphs Eulerian and/or Hamiltonian?

Question 99

Level of difficulty: M

Do the following graphs possess a Hamiltonian path and a Hamiltonian cycle?



Solution question 98

Solution hint: What does Eulerian and Hamiltonian mean?

Solution: Path graphs and star graphs are not Eulerian and not Hamiltonian; cycle graphs are Eulerian and Hamiltonian; wheel graphs ($n \geq 4$) are not Eulerian but Hamiltonian.

Solution question 99

Solution hint: The path contains each node only once.

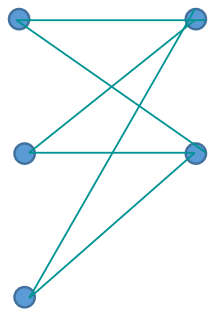
Short solution: Both graphs have a Hamiltonian path.

Detailed solution: In both graphs, there exists an open path that contains each node only once, i.e., an Eulerian path. However, the path cannot be closed.

Question 100

Level of difficulty: D

Show that the graph is not Hamiltonian. Use the result in reference to bipartite graphs.



Solution question 100

Solution hint: If a graph is Hamiltonian with more than two nodes and bipartite, then the two sets of nodes that form the bipartite partitioning are equipotent.

Short solution: Disjoint subsets are not equally powerful.

Detailed solution: The graph is bipartite. Since the disjoint subsets contain three and two nodes and are therefore not equal, the graph cannot be Hamiltonian.

7.3 The Ore and Dirac conditions

Question 101

Level of difficulty: E

Is a graph non-Hamiltonian if it does not satisfy the Ore condition? Give an example.

Question 102

Level of difficulty: E

If a graph satisfies the Dirac condition, does it also satisfy the Ore condition? If a graph satisfies the Ore condition, does it also satisfy the Dirac condition?

Solution question 101

Solution hint: Familiarize yourself with the Ore condition.

Short solution: It does not mean that the graph is not Hamiltonian. Example: C_5

Detailed solution: If a simple graph does not satisfy the Ore condition, this does not mean that it is not a Hamiltonian graph. Thus, C_5 is a simple Hamiltonian graph, but for every two nonadjoint vertices u and v, it holds that $\delta(u) + \delta(v) = 2 + 2 \ngeq n = 5$. This means that the cycle graph C_5 does not satisfy the Ore condition.

Solution question 102

Solution hint: Familiarize yourself with the Ore condition and Dirac condition.

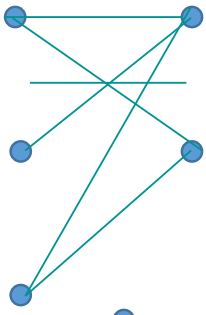
Short solution: Yes; no

Detailed solution: If a graph satisfies the Dirac condition, then it also satisfies the Ore condition. However, if a graph satisfies the Ore condition, then it need not also satisfy the Dirac condition.

Question 103

Level of difficulty: M

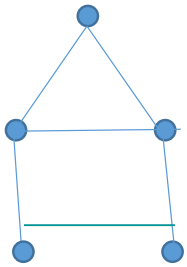
Apply the Ore and Dirac condition to the following graphs. What can you conclude?



Question 104

Level of difficulty: M

Apply the Ore and Dirac condition to the following graphs. What can you conclude?



Solution question 103

Solution hint: Familiarize yourself with the two conditions.

Short solution: Both equations were not fulfilled. Thus, no decision can be made.

Detailed solution: Ore condition: $\delta(u) + \delta(v) = \ngeq n$ with u and v nonadjacent nodes, $\delta(u) + \delta(v) = 3 + 3 = 6$, respectively $\delta(u) + \delta(v) = 2 + 2 = 4 \ngeq 5$. In other words, the Ore condition does not hold for all nonadjacent node pairs. Dirac condition: $\delta(G) \geq n/2$, with $\delta(G)$ minimal degree, $\delta(G) = 2 \ngeq 5/2$. In other words, the Dirac condition is not satisfied. It follows that the Ore and Dirac conditions cannot be used to decide whether the graph is Hamiltonian.

Solution question 104

Solution hint: Familiarize yourself with the two conditions.

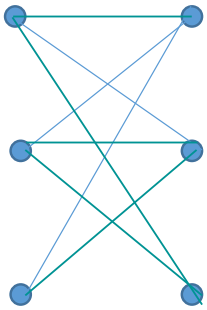
Short solution: Both equations were not fulfilled. Thus, no decision can be made.

Detailed solution: Ore condition: $\delta(u) + \delta(v) = \ngeq n$ with u and v nonadjacent nodes, $\delta(u) + \delta(v) = 3 + 2 = 5$ and $\delta(u) + \delta(v) = 2 + 2 = 4 \ngeq 5$. In other words, the Ore condition does not hold for all nonadjacent node pairs. Dirac condition: $\delta(G) \geq n/2$, with $\delta(G)$ minimal degree, $\delta(G) = 2 \ngeq 5/2$. In other words, the Dirac condition is not satisfied. It follows that the Ore and Dirac conditions cannot be used to decide whether the graph is Hamiltonian. Without proof, let us mention that the graph is Hamiltonian.

Question 105

Level of difficulty: D

Apply the Ore and Dirac condition to the following graph. What can you conclude from this?



Solution question 105

Solution hint: Familiarize yourself with the conditions.

Short solution: Both conditions are satisfied. The graph is Hamiltonian.

Detailed solution: Ore condition: $\delta(u) + \delta(v) = \ngeq n$ with u and v nonadjacent nodes, $\delta(u) + \delta(v) = 3 + 3 = 6 \geq 6$. In other words, the Ore condition is satisfied for all nonadjacent node pairs. Dirac condition: $\delta(G) \geq n/2$, with $\delta(G)$ minimal degree, $\delta(G) = 3 \geq 6/2$. In other words, the Dirac condition is satisfied. It follows that the graph is Hamiltonian.

7.4 Problem of the traveling salesman

Question 106

Level of difficulty: E

What is understood by the traveling salesman problem? Where does it find application in practice? Give an example.

Question 107

Level of difficulty: E

What is the problem with the approach to the problem of the traveling salesman?

Solution question 106

Solution hint: Hamilton cycle problem for weighted graphs

Short solution: Round trip through all nodes minimizing the sum of the weights. Example: electronic circuit boards.

Detailed solution: Closely related to the Hamilton cycle problem is the traveling salesman problem (or TSP). It differs from the Hamilton circle problem only in that the underlying graphs are weighted. The task is to find a round trip through all nodes that minimizes the sum of the distances traveled. So the task is to find a Hamiltonian cycle in a weighted graph such that the sum of the weights is minimal. Example: Electronic circuit board manufacturing.

Solution question 107

Solution hint: Heuristics

Short solution: There is no exact solution.

Detailed solution: There is no exact algorithm that can solve the problem in acceptable time. Of course, it is possible to determine all Hamiltonian cycles, compare the sum of the weights of each, and choose the cycle with the smallest weight. However, as described earlier, a simple graph with n nodes can have up to $(n-1)!/2$ Hamiltonian cycles. This is a lot of possibilities to compare.

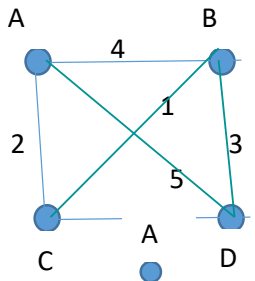
Therefore, a number of efficient approximation algorithms, called heuristics, has been developed in the last decades that can solve the traveling salesman problem

with acceptable accuracy in acceptable time, even with many nodes. However, a heuristic cannot guarantee that the optimal solution can actually be found.

Question 108

Level of difficulty: M

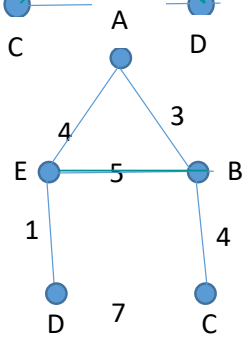
What is the solution to the traveling salesman problem, if the start is in node A?



Question 109

Level of difficulty: M

What is the solution to the traveling salesman problem, if the start is in node A?



Solution question 108

Solution hint: Hamiltonian cycle with minimum weighting

Short solution: e.g., ADBCA with total weight 11

Detailed solution: Find a Hamiltonian cycle with the smallest total weight. A possible solution is ACBCA with total weight 11.

Solution question 109

Solution hint: Hamiltonian cycle with minimum weighting

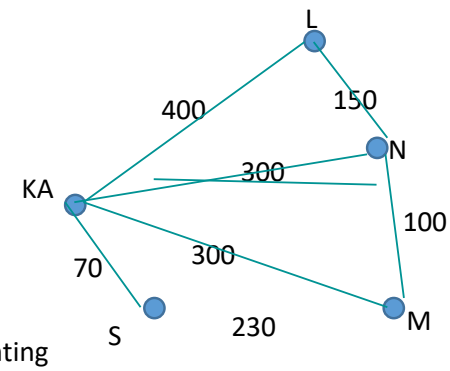
Short solution: AEDCBA with the total weight 19

Detailed solution: Find a Hamiltonian cycle with the smallest total weight. A possible solution is AEDCBA with total weight 19.

Question 110

Level of difficulty: D

A traveling salesman must visit all cities.
Find the optimal circular route if it starts in KA.



Solution question 110

Solution hint: Hamiltonian cycle with minimum weighting

Short solution: KA S M N L KA with the total weight 950.

Detailed solution: Find a Hamiltonian cycle with the smallest total weight. A possible solution is KA S M N L KA with total weight 950.